CSE 465/565
Project#3: 100 points

Outcomes:

- Understand how a Logical language works and understand some of the difficulties
- Implementing rules to solve problems using SWI-Prolog
- Comprehensive testing
- Practice working in a group

Scoring(undergrads/grads):

- (05/05 pts) All responsibilities of group members are clearly explained in detail at the very top of the main code.
- (80/60 pts) Successful implementation of all rules is required.
- (00/20 pts) Successful implementation of extra rules only for graduate students.
- (15/15 pts) The quality of your code will be evaluated based on the percentage of AI tool usage, and how clean and well documented your code is.

Requirements:

- This project MUST be done in groups of up to 4 people.
- Graduate students are required to complete and submit additional work, on top of the regular project requirements. Details about this extra work will be explained later.
- For groups, I suggest starting by dividing the tasks between group members, setting a deadline for each task. Then putting everything together, and save enough time for a **final comprehensive testing**. If you find an issue a day before the deadline, there might not be enough time to go back and fix things. Have a positive attitude, trust each other, and be responsible for doing your part correctly and on time.
- For groups, only one **undergraduate** member should submit the project on Canvas. At the top of your code, you must add comments to include:

1. The names of all group members.
2. All responsibilities of group members must be clearly explained in detail.

- This project is designed to assess your understanding of the material covered in class. You are expected to complete the work yourself, conduct your own research, and use the project as an opportunity to learn more about Scheme.

  I reserve the right to deduct points if a submission contains substantial amounts of code that appear to have been generated **fully** by AI rather than written and understood by the student. Evidence of unauthorized AI use may be treated as a potential academic integrity violation and reviewed according to course and university policies.

  You should be prepared to explain your code, your reasoning, and how your solution works.

★ Group work:

- For smooth and effective work, I suggest the following:
  1. **Stay in communication with your group members**. If you are busy or unable to contribute to the project, communicate this as early as possible so your teammates can plan accordingly and decide how to move forward. Informing your group members that you have been busy only one day before the deadline is not effective communication. It is unprofessional and unfair to those who have been working on the project.
  2. Graduate students are expected to contribute equally to all aspects of the group project. The additional graduate-level requirements are separate responsibilities that must be completed on top of their regular contributions to the group project. **Having extra graduate-level work is not an excuse for taking on fewer responsibilities or contributing less than other group members.**
  3. Please, bring your professional self to the group, have a positive attitude, trust each other, and take responsibility for doing your part correctly and on time.
  4. If you think something should be done differently, speak up and make your case clearly. But if you can't convince the rest of the group, that doesn't give you the right to be upset with them. Be open to the possibility that you might be wrong.
  5. Begin by dividing tasks clearly among group members. Make sure the division is fair and balanced, so everyone has a meaningful role. Be mindful that some problems are harder, and some connected to one another.
  6. Set internal deadlines a couple days before the actual deadline. This gives your group a buffer to handle unexpected issues.
  7. ⇒ If you struggle with starting early or finishing tasks on time, be realistic and honest, and communicate that with the rest of the group. People are different, and there's nothing wrong with that, but not acting professionally and letting your group down is on you. Also, as a group, if you have a group member sharing this with you, please be understanding and set responsibilities and deadlines accordingly and realistically.
  8. If you're not clear about any details in the project, check the discussion board to see if someone has already asked that question. If not, post it on the discussion board or ask your instructor. Please do not assume anything. **You are expected to ask questions. Asking questions is a normal part of all projects, and you need to get used to it.** Once individual parts are completed, get together early to combine everything. This will give your group time for comprehensive testing and final adjustments. Keep in mind that just finishing and testing your part is not the end. You also need to plan for testing each other's work. Ask others to test your solution, and you test someone else's solution.
  9. Communicate regularly and keep everyone updated on your progress. Even a quick check-in can prevent misunderstandings or last-minute surprises.

Instructions:

- Download **p3.pl** and **zipcodes.pl** from canvas, and place both files inside your **Project3** folder. This file contains 10 problems, and your group can work together to solve all of them.
- The instructions inside the p3.pl file describe all problems one by one and include a couple of examples for each.

Graduate Students:

Download the **parse.pl** file. Inside this file you will find the description of what you need to accomplish.

Graduate students are not allowed to work on the graduate part as a group, or share their work with other graduate students. **It must be completed individually**, in addition to their regular group work for the rest of the project. This graduate part is separate from the group project, and having extra work does not give graduate students the right to take on fewer responsibilities in the group work.

⇒ Graduate students must submit **only** their ***parse.pl*** file and **no other files**. Graduate students may not submit the group project on behalf of their group. The group project must be submitted separately by an undergraduate member of the group.

Submission:

1. Make sure that the names of **all group members**, along with a clear and detailed description of each member's responsibilities and contributions, are included as comments at the top of your code.

2. One of the **undergraduate** members can submit on behalf of the whole group.
3. If you are a **graduate** student, you only submit your *parse.pl*.